

EDUCATION

Birla Institute Of Technology And Science, Pilani - Pilani Campus	Aug 2020 - Jun 2024
Bachelor of Engineering in Chemical and Minor in Data Science	CGPA - 8.76

SUBJECTS AND SKILLS

Courses :	Machine Learning, Deep Learning, Artificial Intelligence, Foundation of Data Science, Probability and Statistics, Applied Statistical Method, Derivatives and Risk Management, Principles of Economics, Fundamentals of Finance and Accounting
Technologies :	Python, R, C/C++, SQL, MATLAB, Flutter, Dart, Advanced Excel, Tableau, Snowflake, Amazon RedShift, PowerPoint, Arduino
Libraries :	DeepXDE, Tensorflow, PyTorch, Matplotlib, NumPy, Seaborn, Pandas, SciKitLearn, SciPy, Keras

WORK EXPERIENCE

Manager, AI Discovery and Governance, Tata Steel Limited	Nov 2025 - Present
Enterprise AI Platform: Built the Tata Steel Digital Assistant (TDA) from concept to production in 2 months, reaching 13,000+ active users across 5 group companies in India and the UK (20,000+ queries/day) Data Integration: Integrated 150+ enterprise data sources (100+ databases, 50,000+ documents), cutting policy discovery from 2 business days to under a minute Global Platform Ownership: Own platform management across UK, Thailand, and Singapore (NYK) entities in addition to India, spanning technical, compliance, and stakeholder scope beyond a typical early-career AI role Data Privacy & Compliance: Lead data-privacy and cross-border data transfer compliance discussions for the platform across jurisdictions Next-Gen Architecture: Architecting the next version of TDA on Google's Agent Development Kit (ADK) v2 with improved cross-system connectivity Infrastructure & Vendor Management: Lead contract negotiations with cloud and AI infrastructure partners; developed working fluency in networking and systems fundamentals (MPLS links, infra topology) to support platform scaling Security & Reliability: Delivered enterprise-grade security (RBAC, encryption, VAPT clearance) at 99% uptime; launched a no-code AI agent platform now running 200+ agents across 20 departments	
Manager, AI, Tata Steel Limited	Mar 2025 - Nov 2025
- Delivered an AI-powered development platform used by 700+ developers (React, Flask, GCF, BigQuery, Bedrock) - Built a pipeline converting CXO input into BRDs, FRDs, and deployment-ready technical documentation - Built a project management system tracking INR 1 crore+ in proposed ideas and enabling INR 400 crore+ in organization-wide savings, adopted by 100,000+ people	
Management Trainee, Tata Steel Limited	Aug 2024 - Mar 2025
Rotated across React, C#, R, Python, GCP, SAP, and the Google AI Platform, building the technical foundation for the AI leadership role that followed	
Junior Associate, Indus Insights and Analytical Services Pvt. Ltd	Jan 2024 - Jun 2024
Client Management: Analyzed large datasets for a prominent US-based lender, delivering actionable insights, reducing credit risk by 22% Process Automation: Automated financial reports, cutting processing time by 10 hours (91% decrease), boosting operational efficiency Data Integrity: Implemented logging, quality checks, and flagging in Python, reduced errors by 95% and debugging time by 50% Workflow Optimization: Streamlined SQL data extraction and transformation workflows, increased processing speed by 73% Business Reporting: Created Excel reports, enhancing client understanding of business and improved key performance indicators by 26% Product Analysis: Developed an analysis report for a new product with 5,000+ users, refining features and increased engagement by 18% Dashboarding: Developed 20+ client-facing dashboards on Tableau, enhancing data accessibility and client decision-making capabilities	
IT Intern, Tata Steel Limited	May 2023 - July 2023
Business Insights: Created 10+ interactive visualizations using Tableau and optimized refractory orders to reduce delays by 16% Data Driven Strategy: Analysed over 10 million data points, supporting decision-making process and increasing in strategic accuracy by 24% NLP Innovation: Proposed and received endorsement for a new natural language processing chatbot, expected to enhance internal transparency	
Member of Technical Staff, MASTH Life	May 2022 - Dec 2022
Successfully managed the release process for MASTH and MASTH Guru on Play Store, ensuring smooth deployment - Enhanced app performance, scalability and seamless authentication, resulting in a 30% increase in user engagement and reduced latency. - Coordinated with multiple teams to deliver high-quality updates, reducing app crashes by 40% and boosting overall performance by 20%	

PROJECTS

Image Generation from Text Based Captions using StyleGANs (Report)	
- Generated multiline textual captions from 700 one-hot encoded data points, representing 89 facial features, using a refined word2vec model - Trained a StyleGAN on 700 images and employed a VGG16 model to generate latent spaces, resulting in high-quality image synthesis - Developed a text encoder to produce targeted images, integrating desired facial features from text captions, thereby generating relevant images	
Producing semantic rich latents and high-res images via Diffusion Autoencoders (Report)	
- Produced images from fine-tuned Denoising Diffusion Implicit Models (DDIMs) using noise sampled from normal distribution - Performed linear interpolation on latent space to achieve a smooth morphing between any two given images showcasing different steps - Achieved a Frechet Inception Distance of about 21 from DDIM (lower is better) as compared to 150 from UNET architecture	
Statistical Analysis and Forecasting of Wind Energy	
- Conducted statistical analysis of meteorological data of Rajasthan, including temperature and pressure, to accurately forecast wind speeds - Utilized ADF and KPSS tests to confirm that data is stationary and applied advanced models such as ARMA, ARIMA, SARIMA and LSTM - Evaluated model performance using RMSE, MAE, and MAPE, achieving notable results with SARIMA (82.34% accuracy)	
Development of Convolutional Neural Network (CNN)	
- Developed a Convolutional Neural Network (CNN) from scratch in C++ without utilizing any external libraries or functions - Custom Functions: Created and optimized all functions required for the CNN, including convolution, pooling, and backpropagation	

- Successfully tested the custom-built CNN on the MNIST dataset, achieving an accuracy of 90%, demonstrating the model's effectiveness

PUBLICATIONS

Development of machine learning based model for low-temperature PEM fuel cells ([Research Paper](#))

Mentor : [Dr. Jay Pandey](#)

- Developed a fundamental model of a PEM fuel cell using MATLAB, this model served as a critical foundation for further research and analysis
- Conducted in-depth exploratory data analysis on 20,000+ data points from both theoretical and experimental simulations under various conditions
- Trained and tested Random Forest, Artificial Neural Networks, and Gradient Boosting Trees, achieving an impressive **accuracy rate of 98%**
- Published detailed findings in the esteemed journal "Computers and Chemical Engineering" having cite score as 8.7 and impact factor as 3.9

EXTRA CURRICULAR ACTIVITIES

Member of Student Faculty Council, BITS Pilani

- Facilitated meetings with 8 faculty members and 100+ students to implement academic and student condition improvements across all years

Workshop Organizer - APOGEE, Topic :Using R for Data Analysis and Modelling

- Conducted workshops on "Using R for Data Analysis," training 50+ students and receiving a 95% positive feedback rating

ACHIEVEMENTS

2nd, IEEE Hackathon, APOGEE, BITS Pilani

- Developed an Arduino-based safety solution with fire alarm and environmental controls, winning a Rs 10,000 Arduino kit and GSM SIM Module

3rd, Chem-Intelligence, Fugacity, IIT Kharagpur

- Conducted exploratory data analysis, identifying relevant features and achieving **96% accuracy** on test data using Gradient Boosting Trees

3rd, Bizcubator, Dhurva College of Management Hyderabad

- Pitched an idea of creating a single interface for courses over 30+ different websites and 10000+ courses to rank 3rd amongst 100+ participants

3rd, Qrious, APOGEE, BITS Pilani

- Solved complex puzzles and challenges, showcasing proficiency in analytical thinking, problem-solving, and applying advanced data techniques.

VOLUNTEERING

Project Lead, Pilani Ki Pathshala, Nirmaan Organization (A registered NGO, started by students of BITS Pilani in 2005)

- **Led a team of 13 volunteers** to provide education and facilities to uplift around **40 children** living in Mungda Basti
- Coordinated with 3 different projects to disburse **5 mobile phones** to beneficiaries under Pranayas Scheme allowing them to attend online classes
- Organized a Momo eating competition during our fest APOGEE 2022 with a footfall of more than 1000 people and managed a budget of 1 lakh

Teaching Assistant, PDP - 1, Department of Chemical Engineering

- **Conducting weekly labs** for more than 100, 3rd year undergraduate students on applications of Chemical Engineering on ASPEN Hysys